

Sanjay Dilip

+1 (469) 274-5546 | Dallas, TX

sanjay.dilip3012@gmail.com | <https://www.linkedin.com/in/sanjaydilip/> | <https://sanjay-dilip.github.io>

Data Scientist/ML Engineer specializing in predictive modeling and applied NLP systems. Reached 0.94 AUC on a 1.1M-record credit risk model and 85.0% recall@10 in a hybrid retrieval RAG system deployed across 41 SEC filings.

EDUCATION

The University of Texas at Dallas

Master of Science, Business Analytics & Artificial Intelligence (STEM)

Aug 2024 - May 2026

GPA: 3.69

Coursework: Database Foundations, Advanced Statistics, Predictive Analytics, Econometrics, Machine Learning, NLP

Sri Sai Ram Institute of Technology, Chennai, India

Sep 2020 - Apr 2024

Bachelor of Technology, Artificial Intelligence and Data Science (STEM)

SKILLS

- **Data Analysis & Querying:** Python, SQL, pandas, NumPy, Excel, data cleaning, data validation, exploratory analysis
- **BI & Reporting:** Power BI, Tableau, IBM Cognos, KPI dashboards, DAX, structured reporting
- **Machine Learning, AI & Modeling:** XGBoost, Random Forest, neural networks, CNNs, NLP, PyTorch, forecasting, regression, classification, feature engineering
- **Data Engineering & Tools:** Azure Blob Storage, Azure Data Factory, ETL/ELT pipelines, Parquet, Git, MongoDB
- **Soft Skills:** Stakeholder presentation, cross-functional communication, analytical storytelling

PROFESSIONAL EXPERIENCE

The AI Research Collective

AI/ML Engineer (Remote)

New York, NY

Sep 2026 – present

Community Dreams Foundation

Business Analyst (Remote)

Dallas, TX

Jul 2026 – Sep 2026

- Built a privacy auditing application that detects and redacts sensitive information before public AI tool submission, covering 4 data categories with Gemini-based name detection, fallback handling, and 14 documented test cases.
- Developed AI applications across healthcare and real estate use cases by separating decision logic from LLM responses, validating outputs against source information, and contributing to production automation systems focused on model reliability.

MedTourEasy

Data Analytics Intern

India

Oct 2023 – Nov 2023

- Predicted blood donor return likelihood at 0.78 AUC by engineering RFM and time-since-donation features from patient records into a binary classification model.
- Redirected outreach priorities by building IBM Cognos and Excel dashboards that surfaced collection volume as the highest-variance metric; the team lead used this to reorient the donor contact strategy.

Evolve Robot Lab

AI Intern

Chennai, India

Jan 2023 – Feb 2023

- Designed and delivered a 4-week Deep Learning curriculum for 4 students, running from ML fundamentals through neural network architecture to a CNN image classification capstone.
- Achieved stable gyroscopic movement on a drone by calibrating PID parameters across X, Y, and Z axes, verified through remote control testing.

PROJECTS

Levi's RAG (AI Diligence Copilot) | Python, FastAPI, Next.js, Supabase, Gemini, BM25 + Dense Retrieval

- Reached 85.0% recall@10 on a 60-question evaluation set, up from a 40.0% baseline, by diagnosing and re-embedding retrieval-diluted answer chunks in a hybrid BM25 and dense-vector pipeline across 41 SEC filings.
- Deployed a live FastAPI backend and Next.js frontend where every answer traces to a specific SEC filing sentence and one of four evidence tiers based on source reliability.

NBA Win Probability Engine | Python, scikit-learn, Streamlit, Google Cloud

- Reached 0.7357 AUC predicting NBA game winners before tip-off by training on three full regular seasons and testing on the following season as a true future holdout.
- Deployed a live Streamlit dashboard with automated post-game refreshes by building a Google Cloud Run and Cloud Scheduler pipeline that rebuilds predictions without manual intervention.

Credit Risk Modeling with AMEX Data | Python, XGBoost, SHAP, Neural Networks

- Reached 0.94 AUC on held-out test data by engineering 13-month customer behavior features from a 1.1M-record AMEX development sample, aggregating delinquency, spend, payment, and balance variables across 30, 90, and 180-day windows, then comparing XGBoost against a Neural Network that scored 0.92.
- Quantified the default rate tradeoff (4.1% vs 7.8%) between conservative and aggressive approval thresholds using SHAP-informed XGBoost simulations.

Image Classification Architecture Comparison (CIFAR-10) | PyTorch, CNN, Data Augmentation, CrossEntropyLoss

- Increased test accuracy from 45.5% to 80.73%, a 35.23 percentage point gain, by replacing a fully connected baseline with a three-layer CNN using data augmentation and dropout.
- Identified cat-dog confusion as the model's weakest point by analyzing per-class accuracy and a full confusion matrix, with cat recall lowest at 58.2% against a 90.5% high for ship.

LICENSES AND CERTIFICATIONS:

SnowPro Associate (S126888-251211-SOL) | Microsoft Azure Data Fundamentals (CCFF772D4BC0425B)